

Intelligent Humanoid Airport Concierge: A RAG-Based Voice Interface and Autonomous Navigation System for the Unitree G1

M.A.A. Rafi

Department of Computer Science and Engineering
University of Moratuwa
Moratuwa, Sri Lanka
Email : abdulr.23@cse.mrt.ac.lk

Sulochana Sooriyaarachchi

Department of Computer Science & Engineering
University of Moratuwa
Katubedda, Sri Lanka
Email : sulochanas@cse.mrt.ac.lk

Keywords - humanoid robotics, retrieval-augmented generation, autonomous navigation, LiDAR SLAM, multilingual speech interface

I. INTRODUCTION

Airport terminals need both natural-language guidance and physical wayfinding. Existing systems only partly cover this: chatbots answer questions but cannot escort a passenger, while navigation robots move reliably but understand little natural language. This work integrates both on a Unitree G1 EDU humanoid. A retrieval-augmented voice interface answers passenger queries from an approved knowledge base, and a LiDAR-based navigation stack then guides the passenger to the identified destination. The system targets English, Sinhala and Tamil. It is validated in MuJoCo/ROS 2 simulation, where a spoken request can already send the robot to a pre-saved location.

II. LITERATURE REVIEW

Retrieval-augmented generation (RAG) grounds language-model outputs in an external document store, reducing hallucination and allowing knowledge updates without retraining [1]. The ROS 2 Nav2 stack offers a modular, behaviour-tree-driven planner and controller demonstrated on service robots operating alongside pedestrians [2]. Multilingual transcription remains a bottleneck for lower-resource languages such as Sinhala and Tamil. Whisper generalises well but can trail cloud recognisers on noisy, accented speech [3]. faster-whisper [5] reimplements it to cut inference time and memory. Few prior systems combine RAG conversation, multilingual speech and full-body humanoid navigation in one airport-concierge pipeline.

III. MATERIALS AND METHODS

A. Problem Formulation and Objectives

The target scenario is a passenger asking, for example, “Where is the taxi pickup point?” The robot must transcribe

the request, retrieve approved information, give a grounded spoken answer, identify the destination on its map, and guide the passenger there autonomously. The objectives are to: (1) build a voice interface for the G1; (2) ground answers in approved documents; (3) support English, Sinhala and Tamil; (4) map, localize and navigate to a selected destination, validated in simulation first; and (5) evaluate accuracy, latency, navigation success and computational cost.

B. System Architecture

Fig. 1 shows the two pipelines. In the conversation pipeline, WebRTC voice-activity detection (VAD) segments speech, which is transcribed and passed to a LangChain agent. The agent retrieves context from a FAISS index of approved documents (Gemini embeddings) and a Gemini model generates a grounded spoken answer. If the request names a destination, the agent matches it to a pre-saved map location and forwards it as a Nav2 goal. The goal is planned on a LiDAR-built map, localized through the frame chain odom $\rightarrow$ pelvis $\rightarrow$ mid360_link, and executed as velocity commands on the simulated G1.

C. Conversational RAG Pipeline

Table I lists the components chosen for the conversation pipeline; speech recognition uses Whisper, with faster-whisper [5] as a lighter local backend. RAG was preferred over fine-tuning because the knowledge base can be updated by re-indexing documents without retraining, which matters where airport signage, routes and services change.

D. Localization and Navigation

Indoor maps are built in MuJoCo/ROS 2 simulation from a simulated Livox MID-360 LiDAR, an Intel RealSense D435 depth camera, an IMU and ground-truth odometry (Fig. 2). Given a 2D map and an initial pose, Nav2 plans a global path and converts it into local velocity commands. It continuously updates the pose and obstacle estimate. To

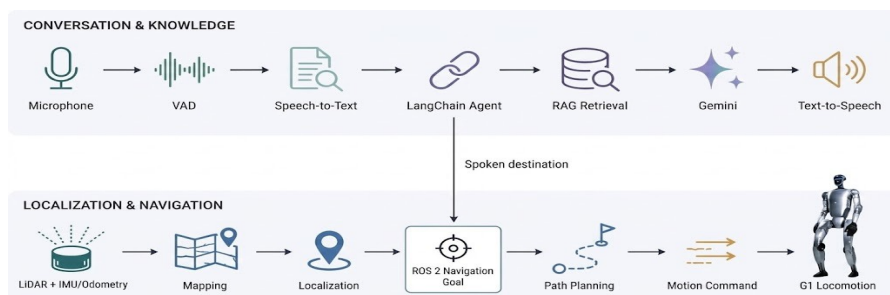


Fig. 1. System architecture: the conversation pipeline (top) passes a spoken destination to the ROS 2 navigation pipeline (bottom).

keep the humanoid stable, velocity limits are set conservatively at about 0.2 m/s linear and 0.3 rad/s angular.

E. Evaluation Method

Conversational grounding was evaluated with RAGAS [6], where an LLM judge scores each question-answer pair on four dimensions: Faithfulness (answer claims supported by the retrieved context), Answer Relevancy (answer addresses the question), Context Precision (relevant chunks ranked near the top) and Context Recall (retrieved context contains the needed information). Both the generator and the judge used Gemini 3.5 Flash-Lite, with $k = 3$ chunks per query and answer-relevancy strictness set to 1 because of free-tier API limits. The test set is $n = 16$ held-out English questions on wayfinding, services and policy. Navigation has so far been assessed qualitatively, by checking that the simulated G1 maps its environment and reaches selected goals. Quantitative navigation metrics (success rate, time to goal, path length) are planned.

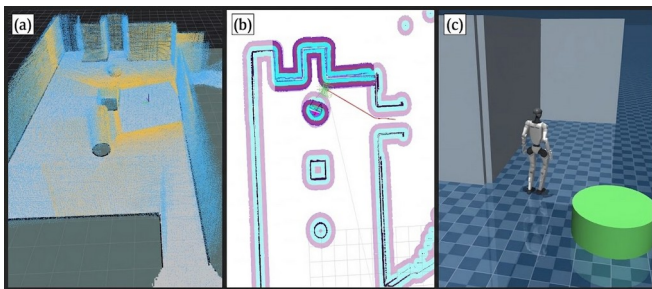


Fig. 2. (a) 3D point cloud visualization in RViz[4]; (b) Nav2 costmap and LiDAR scan in RViz; (c) Unitree G1 Humanoid robot walking in the MuJoCo simulation.

TABLE I. METHOD SELECTION AND JUSTIFICATION

Component	Method	Justification
Speech detection	WebRTC VAD	Lightweight; avoids processing silence
Speech recognition	Whisper / faster-whisper; cloud STT	Multilingual (EN/SI/TA) with a local backup
Retrieval and grounding	Gemini embeddings + FAISS; RAG with Gemini	Efficient search; restricts answers to approved sources
Orchestration	LangChain	Connects conversation, retrieval and navigation
Mapping / navigation	LiDAR SLAM + Nav2	Mature, modular indoor navigation framework

IV. RESULTS AND DISCUSSION

A. Completed Work

The G1 MuJoCo simulation, simulated LiDAR and depth sensing, ROS 2 odometry/TF, voxel mapping, RViz visualization and text-to-speech output are complete. English RAG conversation and autonomous navigation to a selected goal have each been demonstrated end-to-end in simulation. The two are now connected: a spoken request sends the simulated G1 to a pre-saved location.

B. Conversational Evaluation Results

All 16 questions completed with zero evaluation errors (Fig. 3). The pipeline achieved a mean Faithfulness of 0.8979 (± 0.1991), Answer Relevancy of 0.8837 (± 0.1280), Context Precision of 0.8906 (± 0.2167) and Context Recall

of 0.9792 (± 0.0833). The macro-average RAGAS score is 0.9129. Context Recall was the highest and most consistent score, so the FAISS retriever usually captured the needed information. The main weaknesses were capability-related questions, which lowered Faithfulness, and retrieval ranking for some location questions, which lowered Context Precision.

C. Limitations

Sinhala and Tamil recognition is still being tuned. Local Whisper models are less reliable for these languages, and a cloud recogniser improves them. Running perception and simulation together slows navigation. Depth-camera 3D obstacle avoidance is not yet implemented, so the 3D LiDAR may miss certain obstacles. RAGAS scores are LLM-judge estimates rather than ground-truth accuracy, so they are proxies pending correlation with human-annotated judgments. Speech-commanded navigation is limited to pre-saved locations. Evaluation is English-only, and navigation has fewer quantitative measurements.

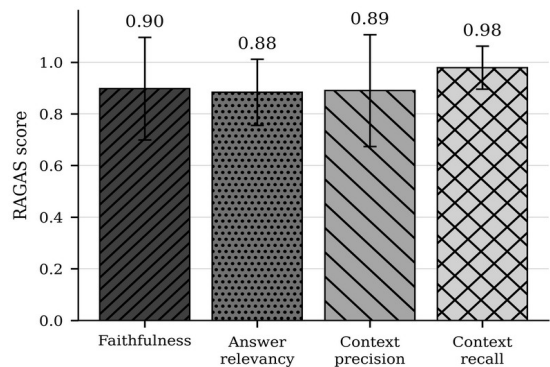


Fig. 3. Mean RAGAS evaluation scores (Faithfulness, Answer Relevancy, Context Precision, Context Recall) over $n = 16$ test questions, scored by an LLM judge (Gemini 3.5 Flash-Lite). Error bars show one standard deviation across questions.

V. CONCLUSION AND FUTURE WORK

A single humanoid platform can combine grounded conversational AI with autonomous indoor navigation for an airport-concierge role. English conversation and speech-commanded navigation to pre-saved locations work in simulation. Future work will add quantitative navigation metrics, extend destinations beyond pre-saved locations, replace ground-truth odometry with LiDAR-inertial localization, improve Sinhala and Tamil recognition, add depth-based 3D obstacle detection, and test the physical G1 with real users.

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